

Deploying the Magic Wand Project

In this document we are going to deploy the pre-trained magic wand project which can recognize gestures of the digits 0-9 by analyzing the time series data produced by the IMU!



Building the Wand

Note that you do not have to build the full magic wand and can simply hold the Arduino (with or without the shield) in your hand. However, you will likely find that when you deploy the example with Pete's pre-trained digit model, since he trained it using a 'wand,' it will be more accurate if you use a 'wand' too.

If you'd like to build a 'wand,' the 'wand' itself can be as simple as a stick. It doesn't need to do anything other than keep the board at its end as you hold the other end and wave it about. For fun, the course staff members have found cheap wands from online retailers and used them, but a simple piece of wood, a ruler, or even a thick piece of cardboard should work just as well.

You should place the board at the end of the wand, with the USB socket facing downwards, towards where you hold it, so that the cable can run down the handle. The sketch is designed to compensate for any rotation of the board around the wand's shaft, so as long as it's parallel to the wand's length, the board's twist won't matter. Use some sticky tape or some other easy-to-remove method to attach the board to the wand, and hold the cable in place along the shaft. The end result should look something like the image below.



<Alt-text: The board attached to a “wand.” The board is taped around the USB cable at the bottom of the board and around the BLE module at the top of the board to a wooden stick.>

If an ASCII-art diagram is more helpful, here's what you should aim for:

```

----
|      |<- Arduino board
|      |
| ( )  |  <- Reset button
|      |
- - - -  <- USB port
  ||
  ||<- Wand
. . . .
  ||
  ||
  ( )

```

Deploying the Magic Wand Project

1. Use a USB cable to connect the Arduino Nano 33 BLE Sense to your machine. You should see the green LED power indicator come on when the board first receives power.
2. Open the magic_wand.ino sketch, which you can find via the File drop-down menu. Navigate, as follows: [File](#) → [Examples](#) → [Harvard_TinyMLx](#) → [magic_wand](#).
3. If you are using a Nano 33 BLE rev2 board, uncomment line 16 (see below). This will load the Arduino_BMI270_BMM150.h IMU library which is compatible with the rev2 board instead of the Arduino_LSM9DS1.h IMU library which is compatible with the rev1 board.

```

// if using Nano 33 BLE rev2, uncomment next line
// #define NAN033_BLE_REV2

```

4. As always, use the Tools drop down menu to select appropriate Port and Board.
 - a. Select the Arduino Nano 33 BLE as the board by going to **Tools → Board: <Current Board Name> → Arduino Mbed OS Boards (nRF52840) → Arduino Nano 33 BLE**. Note that on different operating systems the exact name of the board may vary but/and it should include the word Nano at a minimum. If you do not see that as an option then please go back to Setting up the Software and make sure you have installed the necessary board files.
 - b. Then select the USB Port associated with your board. This will appear differently on Windows, macOS, Linux but will likely indicate 'Arduino Nano 33 BLE" in parenthesis. You can select this by going to **Tools → Port: <Current Port (Board on Port)> → <TBD Based on OS> (Arduino Nano 33 BLE)**. Where <TBD Based on OS> is most likely to come from the list below where <#> indicates some integer number
 - i. Windows → **COM<#>**
 - ii. macOS → **/dev/cu.usbmodem<#>**
 - iii. Linux → **ttUSB<#>** or **ttACM<#>**
 5. Use the rightward arrow to upload / flash the code. Do not be alarmed if you see a series of orange warnings appear in the console. This is expected as we are working with bleeding edge code. You'll know the upload is complete when you red text in the console at the bottom of the IDE that shows 100% upload of the code and a statement that says something like "Done in <#.#> seconds."
- If you receive an error you will see an orange error bar appear and a red error message in the console. Don't worry -- there are many common reasons this may have occurred.
- To help you debug other issues please check out our [FAQ appendix](#) with answers to the most common errors!
6. Now open the serial monitor. The default magic wand will recognize the digits 0-9 based on data recorded by Pete, so your accuracy may vary depending on how close your "gesture handwriting" is to his! **The serial monitor will output first ASCII art showing the gesture you just performed and below it will be the best match label as well as a confidence score between -128 and 127.** The confidence score indicates how strongly the model believes you performed the gesture. Do note that every time you move the board and then stop a new gesture will be processed so don't be surprised to get some odd results as you move the board to prepare for a gesture.

